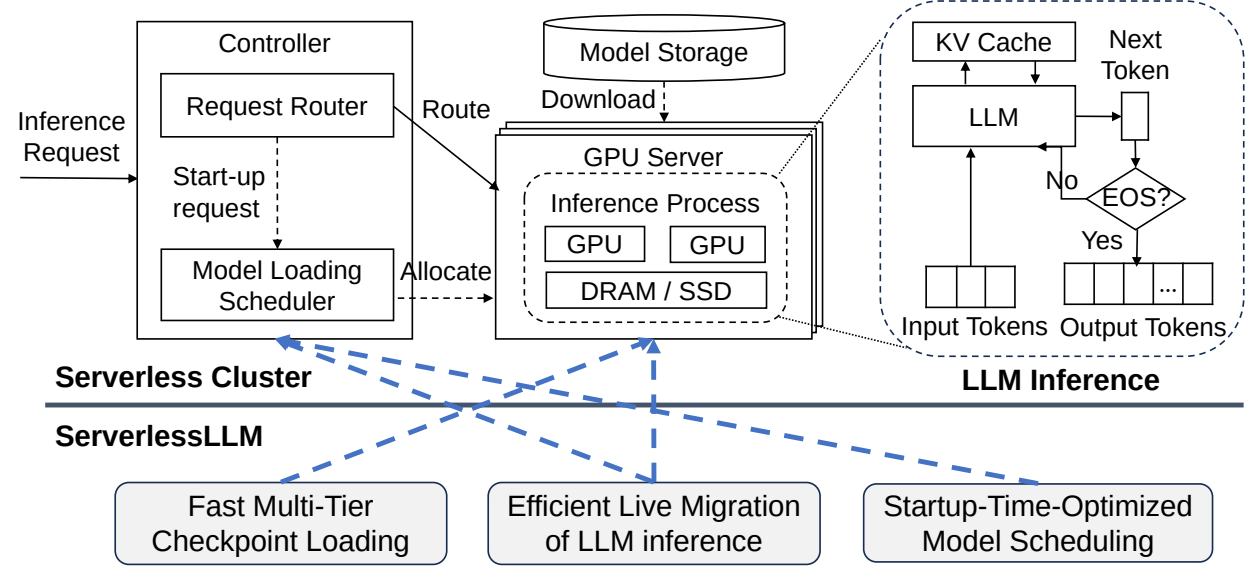

Controller
Request Router
Start-up request
Model Loading Scheduler
Inference Request
Route
Allocate
Model Storage
Download
GPU Server
Inference Process
GPU
GPU
DRAM / SSD
KV Cache
Next Token
LLM
No
EOS?
Yes
Input Tokens
Output Tokens
...
Serverless Cluster
LLM Inference
ServerlessLLM
Fast Multi-Tier Checkpoint Loading
Efficient Live Migration of LLM inference
Startup-Time-Optimized Model Scheduling